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CENTRE
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CANDIDATE
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CHEMISTRY

Grade 12

Paper 2

May 2024

2 hours

Candidates answer on the Question Paper.

Additional Materials: Calculator
 Ruler
 Data Booklet

12CHEM/02

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue, or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You may receive fewer marks if you do not show your working or do not use appropriate units where required.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 100.

Answer all questions in English.

For Examiner's Use

1	
2	
3	
4	
5	
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7	
8	
9	
Total	

This document consists of **17** printed and **3** blank pages.

- 1 Lead can be extracted from galena, lead(II) sulphide. In the first step, lead(II) sulphide is heated in air to form lead(II) oxide and sulphur dioxide.

(a) (i) Write a balanced equation for this reaction.

..... [1]

(ii) Use the Data Booklet to answer this question.

Calculate the volume of sulphur dioxide at room temperature that would be produced from 2.0 kg of galena.

[2]

(b) Sulphur dioxide consists of S=O bonds.

(i) Draw the shape of the sulphur dioxide molecule.

Name the shape produced by the atoms in the molecule **and** suggest a bond angle.

name of shape

bond angle [3]

(ii) Sulphur and silicon each form an oxide of the general formula XO_2 .

Explain, in terms of its structure and bonding, why sulphur dioxide, SO_2 , has a lower melting point than silicon dioxide, SiO_2 .

.....

..... [2]

- (c) In the Contact process, sulphur trioxide, SO_3 , is produced by the oxidation of sulphur dioxide, SO_2 , using the following reaction:



- (i) The reaction is carried out at $400\text{--}450^\circ\text{C}$.
Suggest **two** reasons, why this temperature range is considered a compromise.

.....

.....

.....

..... [2]

- (ii) Write the expression for the equilibrium constant, K_c , for the reaction.

$K_c =$

[1]

- (iii) Predict the effect of increasing the concentration of oxygen on the value of K_c at a constant temperature. Explain your reason.

.....

.....

.....

..... [2]

[Total: 13]

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- 2 A sample of butan-2-ol was heated with acidified potassium dichromate(VI) under reflux, which formed the organic product **X**.

- (a) (i) Give the name of the organic product **X** and state the colour change that would be observed during the reaction.

Name

Colour change [2]

- (ii) Name **and** draw a mechanism for the reaction of organic product **X** with HCN.

[5]

- (b) (i) The reaction in question 2(a)(ii) produces two enantiomers.

Draw 3D structures of these enantiomers.

[2]

- (ii) Explain how two enantiomers can be distinguished from each other.

.....

..... [1]

- (iii) State **one** advantage and **one** disadvantage of using a pure enantiomer in drug production rather than a racemic mixture.

advantage

.....

disadvantage

..... [2]

[Total: 12]

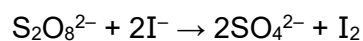
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Question 3 is on the next page.

- 3 (a) (i) Define the term *reduction*.

.....
 [1]

- (ii) Use the Data Booklet to answer this question.
 Calculate the standard cell potential, $E^\ominus_{\text{cell}}$, for the reaction given below.



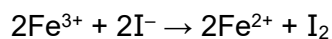
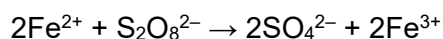
[1]

- (iii) Identify the oxidising ion and the change in oxidation numbers of the oxidised ion.

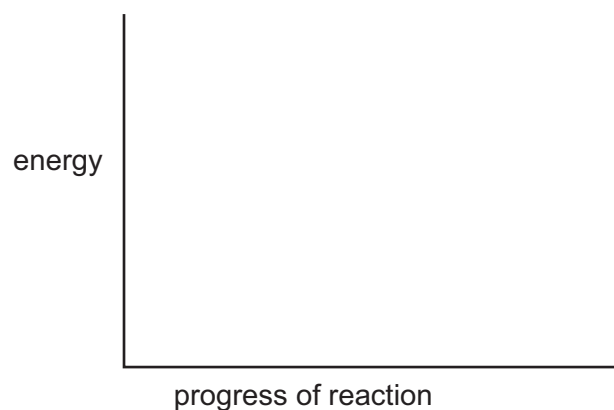
ion

change in oxidation number [1]

- (b) The reaction between peroxodisulphate(VI) ions and iodide ions can be catalysed by iron(II) and iron(III) ions, and these steps are:



- (i) Draw energy level profiles for both the uncatalysed and the catalysed reactions.
 Label each profile.



[2]

- (ii) Explain the reason why the reaction between peroxodisulphate(VI) ions and iodide ions catalysed by $\text{Fe}^{3+}/\text{Fe}^{2+}$ ions is more energetically favourable than the uncatalysed reaction.

.....
.....
.....
..... [2]

- (c) Transition metals and their compounds are used as catalysts in industry and biological systems.

- (i) State **one** property of the transition metal that enables them to act as a catalyst.

.....
..... [1]

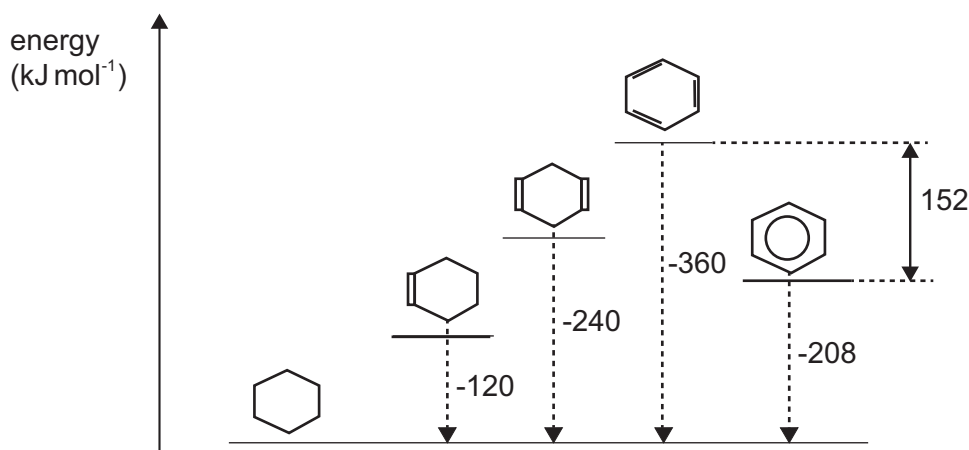
- (ii) Give **one** example of an environmental problem that can be solved by discovering new catalysts in industry.

.....
..... [1]

[Total 9]

- 4 Kekulé structure and π -bonding system are useful models for explaining the bonding and structure of a benzene molecule.

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- (a) (i) Explain the stability of benzene relative to the Kekulé structure. Use the given thermochemical evidence to support your answer.

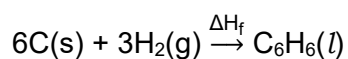
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 [2]

- (ii) The table gives some enthalpy of combustion values.

substance	enthalpy change of combustion /kJ mol ⁻¹
H ₂ (g)	-286
C(s)	-394
C ₆ H ₆ (l)	-3267

Construct a labeled energy cycle to show how these values could be used to calculate the enthalpy change of formation of C₆H₆(l)

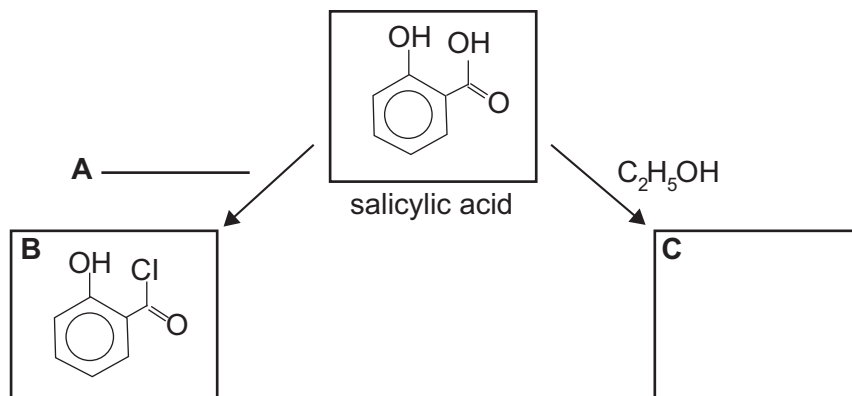


[3]

- (iii) Calculate the enthalpy change of formation, ΔH_f , of C₆H₆(l).

$\Delta H_f =$ kJ mol⁻¹ [2]

(b) Salicylic acid can undergo various reactions as outlined below.



(i) Add the formula of the reagent **A** and the structural formula of the product **C** on the diagram above. [2]

(ii) Compound **B** is also used for making product **C** from ethanol as a salicylic acid. Describe **two** differences between these reactions.

1

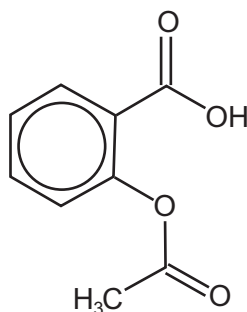
.....

2

..... [2]

(c) Aspirin is an acylation product of salicylic acid.

^1H NMR analysis showed that the spectra of aspirin give different proton environments.



(i) Write the number of proton environments in the aspirin molecule.

..... [1]

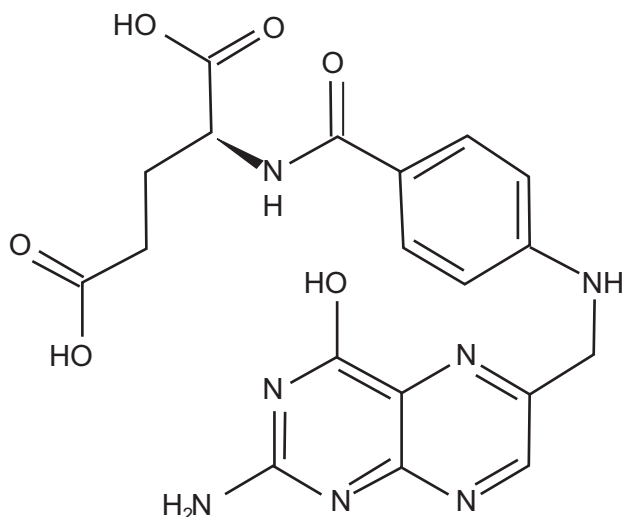
(ii) Write the equation for the reaction of salicylic acid with ethanoic anhydride to produce aspirin.

[1]

[Total 13]

- 5 Vitamin B₉ (C₁₉H₁₉N₇O₆), known as folic acid, has the structure shown below.

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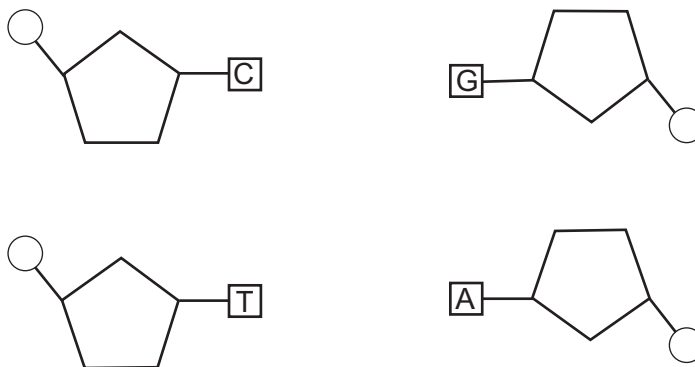
- (a) The soluble sodium salt of folic acid could be used as a drug and may be prepared by adding folic acid to the sodium carbonate.

(i) Identify the functional group responsible for this reaction.

..... [1]

(ii) Folic acid plays an important role in the metabolism of DNA and RNA.

Draw the bonds holding nucleotides together in a part of DNA.



[2]

- (b) Neutrexin is a drug and a competitive inhibitor of enzymes that catalyzes the reduction of intracellular folate compounds.

Explain how neutrexin inhibits the enzyme.

.....

 [2]

(c) Folic acid is a weak acid represented as HA.

(i) Write an equation for the reaction of HA with water.

..... [1]

(ii) Identify the conjugate acid–base pairs in the reaction of HA with water.

.....

..... [1]

(iii) Calculate the pH of a 0.2 mol dm^{-3} solution of folic acid ($K_a = 5 \times 10^{-5} \text{ mol dm}^{-3}$).

[2]

[Total 9]

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6 The Period 3 consists of elements from Na to Ar.

- (a) (i) State the element from Period 3 which is widely used in electronics and has properties important for semiconductor materials.

..... [1]

- (ii) Explain why the first ionisation energies generally increase across the period.

.....
.....
.....
.....
.....
..... [3]

- (iii) Explain why the first ionisation energy of sulphur is less than that of phosphorus.

..... [1]

- (b) (i) Write a balanced equation of the reaction between magnesium and cold water. Include state symbols.

..... [1]

- (ii) Describe **one** observation that would be different if strontium were added to cold water instead of magnesium.

..... [1]

- (iii) State and explain the trend in solubilities in water from magnesium hydroxide, $\text{Mg}(\text{OH})_2$, to barium hydroxide, $\text{Ba}(\text{OH})_2$.

.....

.....

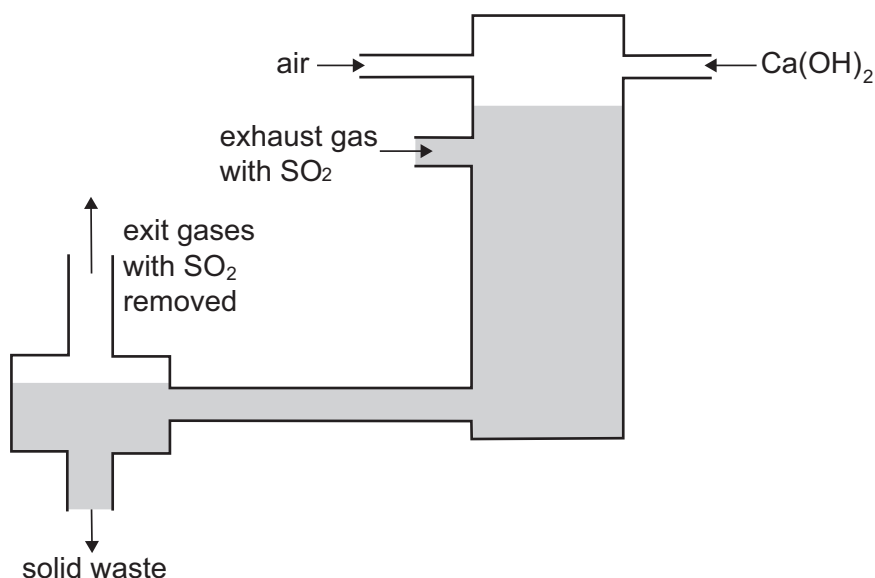
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.....

.....

..... [3]

- (c) Wet scrubbers are used to remove sulphur dioxide (SO_2) and other pollutants from gas streams. One of the most common sorbents is lime ($\text{Ca}(\text{OH})_2$).



- (i) Describe how this method is used to remove sulphur dioxide gases.

.....

..... [1]

- (ii) Sulphur dioxide from gas streams can lead to the development of acid rain. Give **one** environmental effect caused by acid rain.

.....

..... [1]

[Total: 12]

7 2-chloropropane can be obtained by the reaction between propane and chlorine.

(a) (i) Write an equation for this reaction.

..... [1]

(ii) 2-chloropropane can react with hydroxide ions under different conditions.

Name the type of reaction and the required condition for the production of propene from 2-chloropropane.

.....
..... [2]

(b) Propene is used to produce polypropene.

(i) Draw a section of the polymer showing three monomer units joined together.

[1]

(ii) School stacking chairs can be manufactured in a molding process using polypropene.

State **one** physical property of polypropene that makes it a suitable material in furniture manufacturing.

.....
..... [1]

(iii) State **one** environmental problem associated with the disposal of polypropene objects. Explain a reason for this problem.

.....
.....
.....
..... [2]

(c) 2-chloropropane and cyanide ions react together to form methylpropanenitrile.

experiment	$[\text{CN}^-] / \text{mol dm}^{-3}$	$\text{C}_3\text{H}_7\text{Cl} / \text{mol dm}^{-3}$	initial rate / $\text{mol dm}^{-3} \text{s}^{-1}$
1	0.023	0.011	2.7×10^{-6}
2	0.022	0.022	5.5×10^{-6}
3	0.045	0.012	5.5×10^{-6}

- (i) Use the data in the table above to deduce the order of reaction with respect to CN^- and $\text{C}_3\text{H}_7\text{Cl}$, and write the rate expression for the reaction.

.....

 [3]

- (ii) The reaction between 2-chloropropane and CN^- proceeds via both SN_1 and SN_2 mechanisms.

Explain why the rate of the reaction taking place through SN_1 mechanism is proportional to the concentration of $[\text{C}_3\text{H}_7\text{Cl}]$ ions.

.....
 [1]

- (iii) Draw the SN_1 mechanism for the reaction in (c)(ii).

[4]

[Total 15]

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- 8 Buffers play a vital role in many biochemical processes that occur within a narrow range of pH.

(a) (i) Define the term *buffer*.

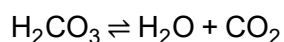
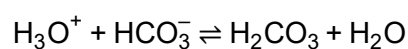
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..... [1]

- (ii) Human blood contains a buffer of carbonic acid, H_2CO_3 , and bicarbonate anion, HCO_3^- , in order to maintain blood pH as shown in the equation below.



Describe how this buffer solution works in controlling the pH of the blood.

.....

.....

.....

..... [2]

- (b) Calculate the pH of the blood buffer solution containing $0.125 \text{ mol dm}^{-3}$ carbonic acid and 0.05 mol dm^{-3} hydrocarbonate ion.
The K_a of carbonic acid is $4.35 \times 10^{-7} \text{ mol dm}^{-3}$.

[4]

[Total: 7]

9 Cobalt is a transition element and forms compounds with oxidation numbers +2 and +3.

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(a) (i) Write the electron configuration of a cobalt atom.

.....
..... [1]

(ii) State the number of unpaired electrons in the Co^{2+} ion.

..... [1]

(b) Cobalt(II) forms complex ions with water ligands.

(i) Define the term *complex ion*.

.....
..... [1]

(ii) Hexaaqua cobalt(II) ions form complex ions with ammonia and EDTA ligands.

Construct the balanced equation between $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and excess ammonia.

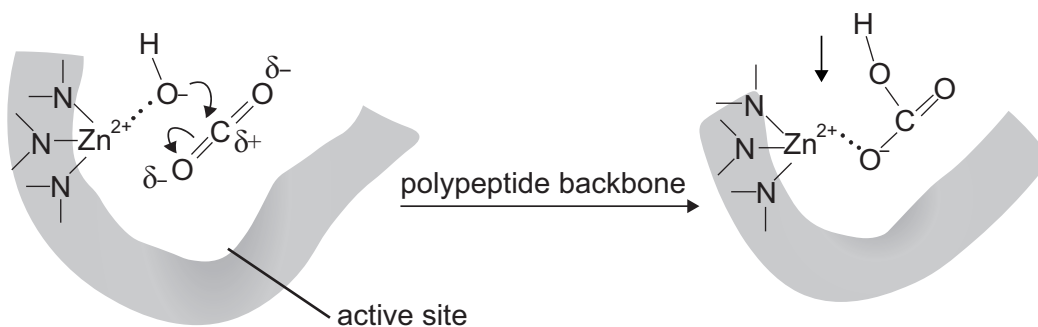
.....
..... [2]

(iii) Explain in terms of Gibbs free energy why the cobalt(II) complex ions with EDTA is more stable than with ammonia.

.....
.....
.....
..... [4]

(c) Zinc is also a d-block element and plays an essential role in biochemical processes.

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With the aid of a diagram, describe the role of Zn^{2+} ions.

..... [1]

[Total 10]

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